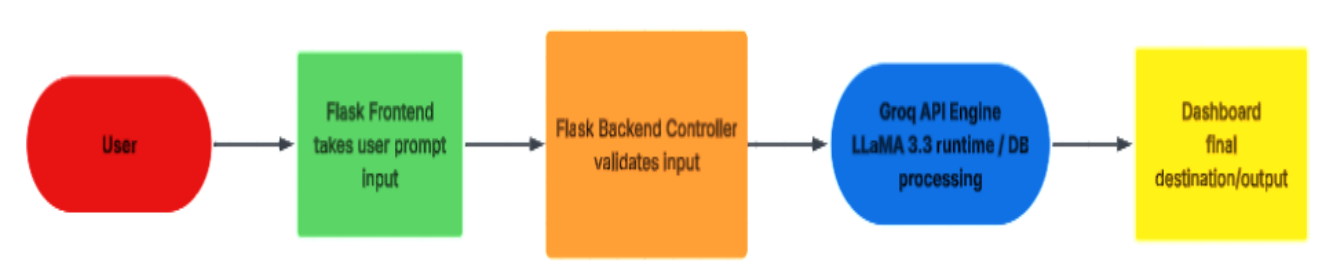


Data Flow Diagram

Date	July7, 2026
Name	Khushboo Meena
Project Name	Caption AI: AI-Powered Social Media Caption Generator

Data Flow Diagram Specification

The following table and structural map describes how data flows through the CaptionAI application, from user input to AI processing and final output.



Source Component (From)	Data Element / Payload Passed	Target Component (To)	Functional Description of Step
User	Product description, selected platform, and preferred tone	Request Handling Module	The user enters the required information and submits the request.
Request Handling Module	Validated user input	Flask Application	The Flask backend receives and processes the user request.
Flask Application	User prompt with API request	Groq API	The backend sends the request to the Groq API for AI content generation.
Groq API	Generated captions, hashtags, and CTAs	Flask Application	The AI model generates the requested content and returns the response to the Flask application.
Flask Application	Formatted response	User Interface	The backend sends the generated content to the frontend for display.

User Interface	Generated captions, hashtags, and CTAs	User	The application displays the final output, allowing the user to review and copy the generated content.
----------------	--	------	--